



Intelligent Control

Session 3: Equilibrium Points and Local Linearization

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Today

1. Foundations & modeling

2. Control & stability

3. Estimation & adaptation

1. Equilibrium points of a nonlinear system
2. Local linearization and the Jacobians
3. What the linearization tells us, and where it stops
4. Simulation lab

Learning objectives

- find the equilibrium points of $\dot{x} = f(x, u)$ by solving $f = 0$;
- build A and B as Jacobians evaluated at a chosen equilibrium;
- decide local stability from those eigenvalues, and recognise the case where they decide nothing;
- design a gain on the linearized model and test it on the true nonlinear plant.

Where we left off

Session 2 gave us a complete theory for

$$\dot{\mathbf{x}} = \mathbf{A}\mathbf{x} + \mathbf{B}u,$$

and Session 1 showed that the pendulum does not fit that template.

Today we build the bridge. The idea is simple and it is the oldest one in control:

near a point where the system is at rest, a differentiable nonlinear model looks like a linear one. So we choose such a point, replace $\mathbf{f}$ by its best linear approximation there, and apply everything from Session 2.

The bridge is not free. We will be equally careful about what it buys and what it costs.

The running example

The actuated pendulum,

$$\ddot{\theta} = -\frac{g}{\ell} \sin \theta - \frac{b}{m\ell^2} \dot{\theta} + \frac{1}{m\ell^2} \tau,$$

with $m = 0.5$ kg, $\ell = 0.3$ m, $b = 0.05$, so $g/\ell = 32.7$, $b/(m\ell^2) = 1.111$, $1/(m\ell^2) = 22.22$.

1. Equilibrium points

The nonlinear system we will study

Throughout this session we consider a nonlinear control system

$$\dot{\mathbf{x}} = \mathbf{f}(\mathbf{x}, \mathbf{u}), \quad \mathbf{x} \in \mathbb{R}^n, \quad \mathbf{u} \in \mathbb{R}^m,$$

where $\mathbf{x}$ is the state, $\mathbf{u}$ is the input, and $\mathbf{f} : \mathbb{R}^n \times \mathbb{R}^m \rightarrow \mathbb{R}^n$ is *continuously differentiable* in a neighbourhood of the operating point, that is $\mathbf{f} \in C^1$.

Our goal is to choose a constant pair $(\mathbf{x}^*, \mathbf{u}^*)$ at which the system can remain at rest, and then approximate the nonlinear dynamics near that pair by a linear model.

Two restrictions, stated once and in force for the whole session:

- $\mathbf{f}$ does not depend on t explicitly, so the system is autonomous;
- we linearize around a *constant* input $\mathbf{u}^*$. Tracking a time-varying reference is Module 2.

Why differentiability matters

The linearized model is the first-order Taylor approximation of $\mathbf{f}$. Continuous differentiability guarantees that the Jacobians vary continuously and that the neglected remainder becomes small *relative to* the deviation from the operating point.

Models with friction discontinuities, backlash or hard saturation are not covered by this argument, even though they are common in practice. We return to that point when the actuator saturates.

What an equilibrium point is

Definition

A pair $(\mathbf{x}^*, \mathbf{u}^*)$ is an **equilibrium** of $\dot{\mathbf{x}} = \mathbf{f}(\mathbf{x}, \mathbf{u})$ when

$$\mathbf{f}(\mathbf{x}^*, \mathbf{u}^*) = \mathbf{0}.$$

The name says it plainly: if the state starts exactly at $\mathbf{x}^*$ and the input is held at $\mathbf{u}^*$, then $\dot{\mathbf{x}} = \mathbf{0}$, so nothing moves. The state stays there forever.

Two remarks that matter later:

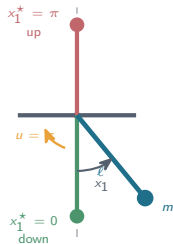
- an equilibrium is a property of the pair, not of the state alone. Change $\mathbf{u}^*$ and the equilibrium moves;
- equilibria are where we *want* the system to sit. Regulation means driving the state to one of them and keeping it there.

Finding them is solving equations

$\mathbf{f}(\mathbf{x}, \mathbf{u}) = \mathbf{0}$ is a system of n algebraic equations, generally nonlinear. It may have one solution, several, infinitely many, or none.

That variety is itself a nonlinear phenomenon: for $\dot{\mathbf{x}} = \mathbf{A}\mathbf{x}$ the equilibria are the null space of $\mathbf{A}$, which is either the origin alone or an entire subspace through it.

The equilibria of the pendulum



x_1 is measured from the **downward** vertical, positive counter-clockwise.

With $x_1 = \theta$, $x_2 = \dot{\theta}$, $u = \tau$,

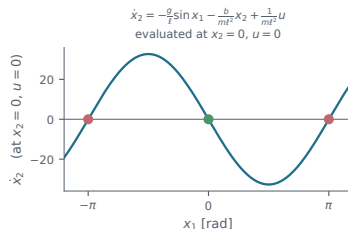
$$f(x, u) = \begin{bmatrix} -\frac{g}{\ell} \sin x_1 - \frac{b}{m\ell^2} x_2 + \frac{1}{m\ell^2} u \\ -\frac{g}{\ell} \sin x_1 - \frac{b}{m\ell^2} x_2 + \frac{1}{m\ell^2} u \end{bmatrix}.$$

Set $u^* = 0$ and solve $f = 0$ row by row.

Step 1. The first row gives $x_2^* = 0$: at rest the velocity must be zero.

Step 2. With $x_2^* = 0$ the second row reduces to $\sin x_1^* = 0$.

Step 3. Therefore $x_1^* = k\pi$: **infinitely many** equilibria, though physically only the two drawn on the left, since angles differing by 2π are the same configuration.



Read the plot

The curve is the second component of f , evaluated where the first already vanishes, $x_2 = 0$, with $u = 0$. The friction term is absent, not omitted.

Its zero crossings at $x_1 = 0, \pm\pi$ are the equilibria, and the *slope* at each crossing decides stability.

The input moves the equilibrium

Nothing forces $u^* = 0$. Keep the torque at a constant u^* and solve again:

$$-\frac{g}{\ell} \sin x_1^* + \frac{1}{m\ell^2} u^* = 0 \quad \Longleftrightarrow \quad \sin x_1^* = \frac{u^*}{mg\ell}.$$

So a constant torque tilts the resting angle: the motor holds the arm away from the vertical, exactly balancing gravity.

The sine takes each value twice per turn, so for $|u^*| < mg\ell$ there are **two** equilibria modulo 2π ,

$$\theta_{\text{down}}^* = \arcsin\left(\frac{u^*}{mg\ell}\right), \quad \theta_{\text{up}}^* = \pi - \theta_{\text{down}}^*,$$

one near hanging down and one near standing up. In all,

- $|u^*| < mg\ell$: two equilibria modulo 2π ,
- $|u^*| = mg\ell$: one degenerate horizontal equilibrium,
- $|u^*| > mg\ell$: no static equilibrium.

This is why we always speak of the pair $(\mathbf{x}^*, \mathbf{u}^*)$: the linearization we are about to build depends on *which* equilibrium we chose.

Numbers

Here $mg\ell = 1.4715 \text{ N m}$.

$u^* = 0.5 \text{ N m}$ gives $\sin x_1^* = 0.34$, hence $\theta_{\text{down}}^* = 0.347 \text{ rad}$ and $\theta_{\text{up}}^* = 2.795 \text{ rad}$, each about 20° away from a vertical.

$u^* = 2 \text{ N m} > mg\ell$ gives no equilibrium at all.

2. Local linearization

The idea, in one dimension first

For a continuously differentiable scalar f , the first-order expansion around x^* is

$$f(x) = \underbrace{f(x^*)}_{=0 \text{ at an equilibrium}} + \left. \frac{df}{dx} \right|_{x^*} (x - x^*) + \underbrace{o(|x - x^*|)}_{\text{smaller than the deviation, near } x^*}.$$

Here $r(x) = o(|x - x^*|)$ means $|r(x)|/|x - x^*| \rightarrow 0$ as $x \rightarrow x^*$.

Two things make this useful:

- the constant term vanishes *because* we expanded around an equilibrium. That is the whole reason we spent the first half of the session finding them;
- what remains, to first order, is a constant times a deviation. That is a linear model.

Defining the deviation $\delta x = x - x^*$ and dropping the remainder,

$$\dot{\delta x} = a \delta x, \quad a = \left. \frac{df}{dx} \right|_{x^*}.$$

The pendulum term

Take $f(\theta) = -\frac{g}{\ell} \sin \theta$ at $\theta^* = 0$:

$$\left. \frac{df}{d\theta} \right|_0 = -\frac{g}{\ell} \cos 0 = -\frac{g}{\ell}.$$

At $\theta^* = \pi$ the cosine flips sign,

$$\left. \frac{df}{d\theta} \right|_{\pi} = -\frac{g}{\ell} \cos \pi = +\frac{g}{\ell}.$$

Same function, opposite slope. That single sign is the difference between hanging and balancing.

The same thing with vectors: the Jacobians

Now $\mathbf{f} : \mathbb{R}^n \times \mathbb{R}^m \rightarrow \mathbb{R}^n$. The derivative becomes a matrix of partial derivatives, one row per component of $\mathbf{f}$, one column per variable.

$$\mathbf{A} = \left. \frac{\partial \mathbf{f}}{\partial \mathbf{x}} \right|_{(\mathbf{x}^*, \mathbf{u}^*)} = \begin{bmatrix} \frac{\partial f_1}{\partial x_1} & \cdots & \frac{\partial f_1}{\partial x_n} \\ \vdots & & \vdots \\ \frac{\partial f_n}{\partial x_1} & \cdots & \frac{\partial f_n}{\partial x_n} \end{bmatrix}, \quad \mathbf{B} = \left. \frac{\partial \mathbf{f}}{\partial \mathbf{u}} \right|_{(\mathbf{x}^*, \mathbf{u}^*)}.$$

Both are **constant** matrices: the partial derivatives are functions, but we evaluate them at the chosen equilibrium and freeze the numbers.

🔗 Two habits worth forming

Write the deviations before anything else,

$$\delta \mathbf{x} = \mathbf{x} - \mathbf{x}^*, \quad \delta \mathbf{u} = \mathbf{u} - \mathbf{u}^*,$$

and note that no derivative is lost in the change of variable, because $\mathbf{x}^*$ is constant:

$$\delta \dot{\mathbf{x}} = \frac{d}{dt} (\mathbf{x} - \mathbf{x}^*) = \dot{\mathbf{x}}.$$

Remember too that $\mathbf{A}$ and $\mathbf{B}$ carry the equilibrium in their definition.

The linearized model

$$\delta \dot{\mathbf{x}} = \mathbf{A} \delta \mathbf{x} + \mathbf{B} \delta \mathbf{u}.$$

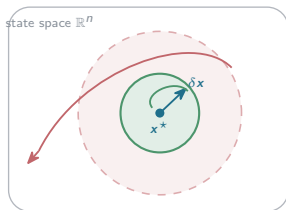
It describes *deviations* from the equilibrium, not the state itself.

Why the remainder is allowed to disappear

The exact statement, with nothing dropped, is

$$\dot{\mathbf{x}} = \mathbf{A} \delta \mathbf{x} + \mathbf{B} \delta \mathbf{u} + \mathbf{r}(\delta \mathbf{x}, \delta \mathbf{u}), \quad \mathbf{r}(\delta \mathbf{x}, \delta \mathbf{u}) = o(\|(\delta \mathbf{x}, \delta \mathbf{u})\|),$$

the same little- o notation as on the previous slide, now for vectors: $\|\mathbf{r}\|/\|(\delta \mathbf{x}, \delta \mathbf{u})\| \rightarrow 0$ as $\|(\delta \mathbf{x}, \delta \mathbf{u})\| \rightarrow 0$.



$\|\delta \mathbf{x}\|$ small: $\mathbf{A}\delta \mathbf{x}$ dominates $\mathbf{r}$ far away: $\mathbf{r}$ can dominate

The theorem promises *some* green disc, but never says how big it is.

Read the limit carefully. It does not merely say that $\mathbf{r}$ goes to zero. It says $\mathbf{r}$ goes to zero *faster than the deviation itself*, so sufficiently close to the equilibrium the first-order term dominates the neglected nonlinear terms.

Consequences for the rest of the session:

- every conclusion drawn from $\mathbf{A}$ is **local**, holding in some neighbourhood of the equilibrium;
- the theorem does not say how big that neighbourhood is;
- far from the equilibrium $\mathbf{r}$ can dominate, and the linear prediction becomes meaningless.

Concretely, for $\sin \theta$

This function is infinitely differentiable, so here we can say more than $o(|\theta|)$:

$$\sin \theta - \theta = -\frac{\theta^3}{6} + \mathcal{O}(\theta^5).$$

The cubic leading order is a property of *this* function, not of the general C^1 result, which promises only $r(\theta) = o(|\theta|)$.

At $\theta = 0.2$: 0.001 33, about 0.7% of θ .
At $\theta = 1.5$: $|\sin 1.5 - 1.5| = 0.5025$, about one third of θ .

Linearizing the pendulum, step by step

The two components are $f_1 = \dot{x}_2$ and $f_2 = -\frac{g}{\ell} \sin x_1 - \frac{b}{m\ell^2} x_2 + \frac{1}{m\ell^2} u$.

Step 1. Write the Jacobians as arrays of partial derivatives, one row per component of $\mathbf{f}$ and one column per variable, and then differentiate each entry:

$$\mathbf{A} = \frac{\partial \mathbf{f}}{\partial \mathbf{x}} = \begin{bmatrix} \frac{\partial f_1}{\partial x_1} & \frac{\partial f_1}{\partial x_2} \\ \frac{\partial f_2}{\partial x_1} & \frac{\partial f_2}{\partial x_2} \end{bmatrix} = \begin{bmatrix} \frac{\partial}{\partial x_1}(\dot{x}_2) & \frac{\partial}{\partial x_2}(\dot{x}_2) \\ \frac{\partial}{\partial x_1}\left(-\frac{g}{\ell} \sin x_1 - \frac{b}{m\ell^2} x_2 + \frac{1}{m\ell^2} u\right) & \frac{\partial}{\partial x_2}\left(-\frac{g}{\ell} \sin x_1 - \frac{b}{m\ell^2} x_2 + \frac{1}{m\ell^2} u\right) \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ -\frac{g}{\ell} \cos x_1 & -\frac{b}{m\ell^2} \end{bmatrix},$$
$$\mathbf{B} = \frac{\partial \mathbf{f}}{\partial u} = \begin{bmatrix} \frac{\partial f_1}{\partial u} \\ \frac{\partial f_2}{\partial u} \end{bmatrix} = \begin{bmatrix} \frac{\partial}{\partial u}(\dot{x}_2) \\ \frac{\partial}{\partial u}\left(-\frac{g}{\ell} \sin x_1 - \frac{b}{m\ell^2} x_2 + \frac{1}{m\ell^2} u\right) \end{bmatrix} = \begin{bmatrix} 0 \\ \frac{1}{m\ell^2} \end{bmatrix}.$$

Step 2. Evaluate at the chosen equilibrium, which freezes the numbers:

$$\mathbf{A} = \begin{bmatrix} 0 & 1 \\ -\frac{g}{\ell} \cos x_1^* & -\frac{b}{m\ell^2} \end{bmatrix}.$$

Only $\cos x_1^*$ changes, taking the value +1 hanging down and -1 standing up.

One symbol carries everything

Three of the four entries of $\mathbf{A}$, and both entries of $\mathbf{B}$, do not depend on where we linearize. The entire difference between the two equilibria sits in $\cos x_1^*$.

That is unusual and convenient. In general every entry of $\mathbf{A}$ changes with the equilibrium.

The two linearizations side by side

🌸 Hanging down, $\theta^* = 0$

$$\mathbf{A}_{\text{down}} = \begin{bmatrix} 0 & 1 \\ -32.7 & -1.111 \end{bmatrix}$$

Characteristic polynomial $\lambda^2 + 1.111\lambda + 32.7$, so

$$\lambda = -0.556 \pm j 5.691.$$

A complex pair with negative real part: **stable focus**. In the language of Session 2, $\omega_n = 5.718$ rad/s and $\zeta = 0.097$, a very lightly damped oscillation.

🌸 Standing up, $\theta^* = \pi$

$$\mathbf{A}_{\text{up}} = \begin{bmatrix} 0 & 1 \\ +32.7 & -1.111 \end{bmatrix}$$

Characteristic polynomial $\lambda^2 + 1.111\lambda - 32.7$, so

$$\lambda = +5.190, \quad \lambda = -6.301.$$

One positive, one negative: **saddle**. The unstable mode grows like $e^{5.19t}$, which doubles a small error every 0.13 s.

One sign flip in a single entry turns a stable focus into a saddle. Both matrices come from the same physical pendulum, differing only in where we chose to sit.

3. What it tells us, and where it stops

The theorem, stated carefully

Lyapunov's indirect method

Let $\mathbf{x}^*$ be an equilibrium of $\dot{\mathbf{x}} = \mathbf{f}(\mathbf{x})$ and let $\mathbf{A}$ be the Jacobian there.

- If every eigenvalue of $\mathbf{A}$ has $\text{Re } \lambda_i < 0$, the equilibrium is **locally** exponentially stable, and hence locally asymptotically stable.
- If at least one eigenvalue has $\text{Re } \lambda_i > 0$, it is unstable.
- If some eigenvalue has $\text{Re } \lambda_i = 0$ and none is positive, the test is **inconclusive**.

The theorem is stated for an autonomous $\dot{\mathbf{x}} = \mathbf{f}(\mathbf{x})$, while we started from $\dot{\mathbf{x}} = \mathbf{f}(\mathbf{x}, \mathbf{u})$. Apply it to the autonomous vector field obtained after holding $\mathbf{u} = \mathbf{u}^*$ constant, or after substituting a state-feedback law $\mathbf{u} = \boldsymbol{\kappa}(\mathbf{x})$, which is exactly what the design section does.

Compare this with the linear test of Session 2 and notice exactly two words changed, both of them expensive:

- *locally*, where the linear statement was global;
- *inconclusive*, where the linear statement still had something to say.

Everything else, the eigenvalue computation itself, is identical to last session. *Khalil, 2002, Thm. 4.7; Slotine & Li, 1991, §3.3.*

The inconclusive case is not a technicality

Consider two scalar systems,

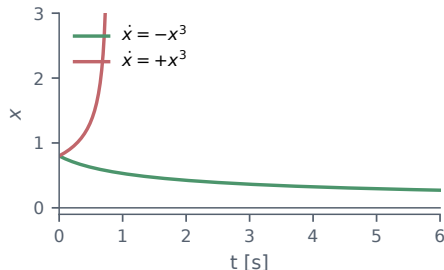
$$\dot{x} = -x^3 \quad \text{and} \quad \dot{x} = +x^3.$$

Both have an equilibrium at $x^* = 0$, and both have the same Jacobian there:

$$\left. \frac{d}{dx} (\pm x^3) \right|_0 = \pm 3x^2|_0 = 0.$$

So $\lambda = 0$ in both cases and the linearization is *the same*: $\delta\dot{x} = 0$. Yet the first converges to the origin and the second escapes to infinity in finite time.

The linearization threw away precisely the term that decides the answer.



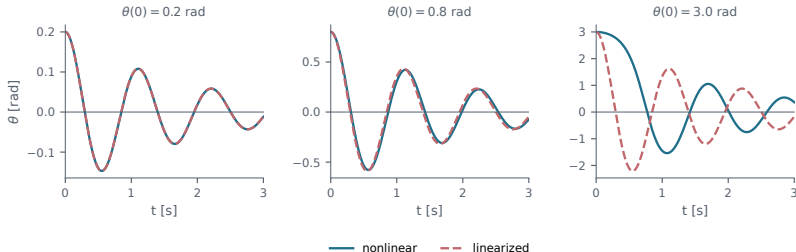
What to take from this

When the linearization gives $\text{Re } \lambda = 0$, it is not being coy. There genuinely is no answer at first order, and the higher-order terms decide.

Module 2 handles exactly these cases, because a Lyapunov function sees $-x^3$ and $+x^3$ as completely different objects.

How local is local?

Same pendulum, released from rest, no torque. The blue curve is the true nonlinear response, the dashed one is the linearized model $\dot{\delta\mathbf{x}} = \mathbf{A}_{\text{down}}\delta\mathbf{x}$.



Measure the disagreement with the same quantity the lab uses,

$$E(\theta_0) = \frac{\max_{0 \leq t \leq 1} |\theta_{\text{nl}}(t) - \theta_{\text{lin}}(t)|}{|\theta_0|}.$$

At $\theta(0) = 0.2$ it is only 0.54%. At 0.8 it grows to 8.59%. At 3.0, close to upright, it exceeds 100%: both still oscillate, but with completely different amplitude and timing, so the linear model carries no quantitative information there.

The honest summary

The theorem promises a neighbourhood but never says how large. Only simulation or a Lyapunov argument gives a number.

A useful habit: never report a linear design without showing the angle at which it stops matching.

4. Designing on the linearization

Balancing the pendulum upright

The recipe is now mechanical, and every step is something we already know how to do.

Step 1. Choose the equilibrium: $\theta^* = \pi$, $\mathbf{u}^* = 0$.

Step 2. Linearize there, which we did two slides ago:

$$\mathbf{A} = \begin{bmatrix} 0 & 1 \\ 32.7 & -1.111 \end{bmatrix}, \quad \mathbf{B} = \begin{bmatrix} 0 \\ 22.22 \end{bmatrix}.$$

Step 3. Place the closed-loop poles with $\delta \mathbf{u} = -\mathbf{K} \delta \mathbf{x}$. Asking for $\omega_n = 6$ rad/s, $\zeta = 0.7$ means $\lambda^2 + 8.4\lambda + 36$. The bottom row of $\mathbf{A} - \mathbf{BK}$ is $[32.7 - 22.22k_1 \quad -1.111 - 22.22k_2]$, so

$$32.7 - 22.22 k_1 = -36, \quad -1.111 - 22.22 k_2 = -8.4,$$

$$k_1 = \frac{36 + 32.7}{22.22} = 3.092, \quad k_2 = \frac{8.4 - 1.111}{22.22} = 0.328.$$

Step 4. Apply it to the *real* plant, undoing the deviation:

$$\tau = -k_1 (\theta - \pi) - k_2 \dot{\theta}.$$

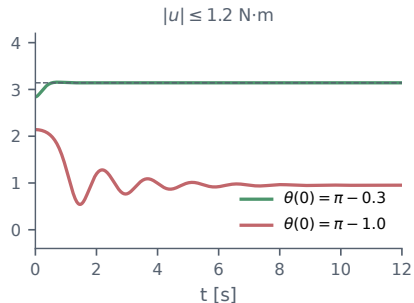
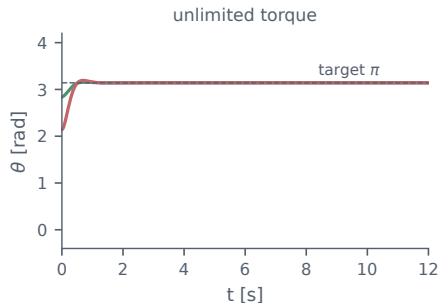
Check the arithmetic

With those gains the closed-loop eigenvalues come out $\lambda = -4.2 \pm j 4.285$, which is exactly $\omega_n = 6$, $\zeta = 0.7$.

What is guaranteed

The theorem guarantees that the *nonlinear* closed loop is locally asymptotically stable at $\theta = \pi$. Nothing more. How far from upright it still works is not part of the promise.

Does it work on the true plant?



Left, with unlimited torque, the design recovers the upright position from these relatively large offsets, well beyond the local behaviour the theorem guarantees. Note that the theorem asserts that *some* neighbourhood works but never produces one, so there is no explicit region to compare against; and this simulation does not certify the full region of attraction either.

Right, the same gain with the torque limited to 1.2 N m. From $\pi - 0.3$ it still balances. From $\pi - 1.0$ it fails to recover the upright position and settles at a *tilted* equilibrium created by the saturated torque.

Saturation changes the equilibrium set

Near the upright equilibrium the controller is unsaturated, so the local linearization remains valid. The saturated behaviour must instead be studied from the piecewise nonlinear closed loop.

Holding the pendulum statically at the horizontal requires

$$mg\ell = 1.4715 \text{ N m.}$$

A smaller actuator cannot balance it there, although the pendulum may still cross the horizontal using kinetic energy.

While the input remains saturated at $u = 1.2 \text{ N m}$, an equilibrium must satisfy

$$mg\ell \sin \theta^* = 1.2.$$

Thus, modulo 2π ,

$$\theta_{\text{lower}}^* = \arcsin\left(\frac{1.2}{1.4715}\right) = 0.954 \text{ rad}, \quad \theta_{\text{upper}}^* = \pi - \theta_{\text{lower}}^* = 2.188 \text{ rad}.$$

The lower equilibrium is locally stable within this saturated branch and the upper one is unstable, so the failed trajectory settles at $\theta = 0.954 \text{ rad}$.

The lesson

Saturation does not merely slow the response: it changes the closed-loop equilibrium set. The region of attraction depends on the nonlinear dynamics, the controller and the actuator limits.

5. Simulation lab

Lab 3A: build the two local models

This code snippet implements the nonlinear pendulum model and the Jacobians derived in class.

The function

$$A_lin(\theta^*)$$

evaluates the state Jacobian at a chosen equilibrium, while B is constant because the input enters the model linearly.

Evaluating

$$\theta^* = 0 \quad \text{and} \quad \theta^* = \pi$$

produces the two local models and their eigenvalues: a **stable focus** and an **unstable saddle**.

Key point: the nonlinear plant is the same; changing the equilibrium changes A and therefore the local stability.

```
m, l, b, g = 0.5, 0.3, 0.05, 9.81
GL, BM = g/l, b/(m*l**2)
IM = 1/(m*l**2)

def f(t, x, u):
    th, om = x
    dom = (-GL*np.sin(th)
           - BM*om + IM*u)
    return [om, dom]

def A_lin(th_star):
    return np.array([
        [0., 1.],
        [-GL*np.cos(th_star), -BM]
    ])

B = np.array([[0.], [IM]])

for th_star in (0.0, np.pi):
    A = A_lin(th_star)
    print(
        th_star,
        np.linalg.eigvals(A)
    )
```

Lab 3A: how good is the linear approximation?

1. Reproduce both linearizations and their eigenvalues. Confirm the stable focus at $\theta^* = 0$ and the saddle at $\theta^* = \pi$.
2. Simulate the nonlinear and the linearized models from

$$\theta(0) = 0.2, 0.8, 3.0, \quad \dot{\theta}(0) = 0, \quad u = 0,$$

and for each case compute

$$E(\theta_0) = \frac{\max_{0 \leq t \leq 1} |\theta_{\text{nl}}(t) - \theta_{\text{lin}}(t)|}{|\theta_0|}.$$

3. By bisection, find the largest $\theta_0 \in (0, \pi]$ with $E(\theta_0) \leq 0.05$. The interval is open at 0 because E divides by $|\theta_0|$; equivalently, set $E(0) = 0$ by convention. Report it as an **empirical validity limit** for this family of initial conditions, this one-second horizon and this metric. It is not the full neighbourhood in which the linearization is valid.

Lab 3B: upright control and saturation

- Design K for the upright equilibrium with $\omega_n = 6 \text{ rad/s}$, $\zeta = 0.7$, and verify the closed-loop eigenvalues.
- Apply it to the nonlinear plant from $\theta(0) = \pi - \Delta$, $\dot{\theta}(0) = 0$, and search $\Delta \in [0, \pi]$ for the largest offset it recovers from. Count a run as successful when, during the *final second* of a 12 s simulation,

$$\max_{11 \leq t \leq 12} |\theta(t) - \pi| < 0.01, \quad \max_{11 \leq t \leq 12} |\dot{\theta}(t)| < 0.01.$$

Testing the whole last second, not just the final instant, avoids accepting a trajectory that merely passes near the equilibrium at $t = T$. Do this for $u_{\max} = 1.2 \text{ N m}$ and $u_{\max} = 3 \text{ N m}$, and put both results in a table.

- For a run that fails, report the angle at which it settles and explain where that equilibrium comes from.

Lab 3A and 3B together are Homework 3.

Lab 3B: the closed loop in code

This code snippet starts items 5 and 6. It closes the loop around the *same* f of the previous code snippet: the model is never modified, only the input it receives.

Lines 3 to 5. K from item 4, $UMAX$ the actuator limit, $XSTAR$ the upright equilibrium x^* .

Lines 7 to 11. $closed(t,x)$ performs, in order, the four operations of the control law:

$$\delta x = x - x^*, \quad u = -K \delta x, \quad u \leftarrow \text{sat}(u), \quad \dot{x} = f(x, u),$$

$np.clip$ is the saturation, the only nonlinearity we added ourselves.

Lines 13 to 20. One run, offset $\Delta = 0.3$. For item 5 wrap it in a search over Δ , and apply the success test to the *last second* of sol , not to its final point.

```
from scipy.integrate import solve_ivp

K = np.array([[3.0915, 0.3280]])
UMAX = 1.2
XSTAR = np.array([np.pi, 0.0])

def closed(t, x):
    e = x - XSTAR
    u = (-K @ e).item()
    u = np.clip(u, -UMAX, UMAX)
    return f(t, x, u)

x0 = [np.pi - 0.3, 0.0]

sol = solve_ivp(
    closed,
    [0, 12],
    x0,
    max_step=0.005
)
```

Hint for item 5: compare $UMAX$ against $mg\ell = 1.4715 \text{ N m}$.

Wrap-up and next session

Today

- equilibria solve $\mathbf{f}(\mathbf{x}^*, \mathbf{u}^*) = \mathbf{0}$, and a nonlinear system may have many, or none;
- the input moves them, so an equilibrium is a property of the pair;
- $\mathbf{A}$ and $\mathbf{B}$ are Jacobians frozen at the chosen equilibrium, and they act on deviations;
- negative real parts give *local* asymptotic stability, positive ones give instability, zero gives nothing.

Next session: Module 2 begins

- regulation and tracking stated as problems, not as recipes;
- the error dynamics, which is where control design actually happens;
- why we need a theory that does not depend on being close to an equilibrium.

On the two open questions of today: Lyapunov theory provides tools to analyse the inconclusive cases, and to *certify estimates* of a region of attraction. It does not hand over the exact or largest such region.

Homework 3: due before the next session

Complete items 1 to 6 of Lab 3 and submit a **single PDF** containing the requested plots, the relevant code and brief explanations. Upload the PDF to the *Homework 3* assignment on Blackboard by the posted deadline. Submissions by email will not be accepted.